

PRAGYA

AI-Powered Competency & Personalized Learning Platform

SIH26101 | Ministry of Statistics & Programme Implementation (MoSPI)

Project north star

Help each official understand what their role requires, identify the most important competency gaps, learn through the right ecosystem resources, prove improvement, and continuously receive a better next step.

This document is the team's master source of truth for product design, feature scope, UI, AI/ML architecture, data, integration approach, security, prototype boundaries, development order and SIH presentation. It separates official SIH requirements from our proposed enhancements.

1. Project identity

Item	Final decision
Problem Statement	SIH26101
Product Name	PRAGYA
Product category	Software
Theme	Smart Education
Target context	India's Official Statistical System
Primary users	Employees / officials
Secondary users	Administrators, training managers, trainers
Core design principle	Competency first, learning second
Hero differentiator	Closed-loop, evidence-backed competency intelligence
UI theme	Dark, colourful, futuristic, neon animated-gradient visual

	language
Prototype priority	Working end-to-end core loop before advanced breadth

2. Problem in simple language

The problem is not simply that employees need more courses. The problem is that employees have different roles, different competency levels and different learning needs. A learning platform should therefore know what an employee's job requires, estimate the employee's current capability from evidence, identify the gap, recommend the most relevant learning, assess the result and then update the next recommendation.

Simple analogy

An ordinary LMS is like a library. PRAGYA is like a career coach + assessment engine + training planner + workforce intelligence layer, connected to the learning ecosystem.

3. Why we need two dashboards

SIH26101 explicitly calls for interactive dashboards for both Learner and Administrator. [S1] They are not two unrelated systems. They are two views of the same competency and learning data, designed for different decisions.

Learner / Employee Dashboard	Administrator Dashboard
Question: What should I learn next?	Question: What does our workforce need next?
Scope: one employee	Scope: organization / department / role
Shows my competency profile	Shows workforce competency distribution
Shows my skill gaps	Shows common and department-wise gaps
Shows my learning path	Shows training demand and effectiveness
Shows my assessments and progress	Shows workforce training outcomes
Lets me use AI learning assistant	Supports capacity-building decisions
Updates my skill profile	Aggregates evidence into workforce analytics

One platform, two views

Employee actions generate evidence in the shared data layer. The employee sees personal outcomes; authorized administrators see aggregated or permitted workforce intelligence. This creates the chain: individual learning -> organizational intelligence.

4. Complete PRAGYA functional flow

The product flow below is the main story we should build and demonstrate.

1. Login / identity
2. Employee profile
3. Role requirements
4. Competency assessment
5. Evidence-backed skill profile
6. Skill-gap analysis
7. Gap priority
8. Personalized learning path
9. iGOT + NSSTA/TPAC options
10. AI learning assistant / RAG
11. Interactive virtual lab
12. AI-generated MCQs
13. Adaptive assessment
14. Performance analysis
15. Competency recalibration
16. Updated recommendations
17. Administrator workforce analytics
18. Emerging skills / future capacity planning

5. Feature architecture

Feature system	What we build	How useful	Requirement status
Identity & profile	Government-style login, SSO-ready identity, role, department, designation, assignment, education, experience, previous training, target role.	Gives the system the context required for personalization.	SIH requirement in detailed description. [S1]
Competency intelligence	33 SIH-listed competency labels across Statistical, Technical, Digital Governance and Behavioural/Managerial domains; evidence-backed scoring and confidence.	Creates the learner capability model.	Core SIH requirement + proposed evidence model. [S1]
Skill-gap engine	Current vs target competency with criticality, task relevance and recency.	Turns assessment into an actionable priority list.	Core SIH requirement. [S1]
Recommendation engine	Hybrid rules + semantic embeddings + metadata ranking.	Chooses the most relevant learning instead of showing a generic catalogue.	Core SIH personalization requirement; hybrid method is our enhancement. [S1]

iGOT integration layer	Course discovery, metadata, enrollment/completion/history adapter; mock provider for prototype.	Connects PRAGYA recommendations to the iGOT ecosystem without inventing live credentials.	SIH requirement. [S1]
NSSTA/TPAC layer	Training-programme catalogue with competencies, eligibility, schedule, duration and source.	Brings structured institutional training into the same learning plan.	SIH requirement. [S1]
AI learning assistant	Profile-aware RAG assistant with source citations, hints, examples, practice prompts and weak-topic context.	Provides course-specific support while learning.	SIH detailed capability; our competency-aware design is an enhancement. [S1]
Material ingestion	PDF/PPT/video upload, extraction/transcription, chunking, embeddings and source metadata.	Turns learning material into searchable knowledge and assessment input.	SIH requirement for uploaded learning materials. [S1]
Quiz / MCQ generator	Concept extraction, competency mapping, question generation, distractors, difficulty, validation, duplicate detection, source links.	Automates assessment creation while keeping it traceable.	SIH requirement + validation layer is our enhancement. [S1]
Adaptive assessment	Difficulty changes based on recent performance.	Measures knowledge efficiently and supports personalized learning.	SIH detailed capability. [S1]
Virtual lab	One high-quality Sampling Lab using synthetic data.	Provides hands-on practice instead of passive content consumption.	SIH detailed capability; one lab is our implementation choice. [S1]
Multilingual learning	English/Hindi/Telugu for selected explanations/UI/content; internal competency IDs stay canonical.	Improves accessibility.	SIH detailed capability. [S1]
Learner dashboard	Competency, gaps, learning path, hours, progress, assessments, history.	Lets one employee decide what to do next.	Explicit dashboard requirement. [S1]
Administrator dashboard	Workforce competencies, distribution, training effectiveness, emerging skills, predictive analytics.	Lets training managers plan organization-wide capability development.	Explicit dashboard requirement. [S1]
Security	RBAC, secure uploads, audit logs, data minimization, AI prompt-injection controls, SSO-ready architecture.	Protects sensitive workforce and learning data.	SIH security/privacy requirements + our implementation. [S1]

6. Competency framework

Domain	Competencies
Statistical	Survey Design; Sampling; National Accounts; Price Statistics;

	Labour Statistics; Agricultural Statistics; Industrial Statistics; SDG Indicators; Metadata Standards; Data Quality Frameworks
Technical	Python; R; SQL; Stata; SPSS; SAS; GIS; Data Visualization; AI/ML; Cloud Computing; APIs; Open Data
Digital Governance	Cybersecurity; Data Privacy; Digital Signatures; Government Cloud; Digital Public Infrastructure
Behavioural / Managerial	Leadership; Communication; Project Management; Ethics; Decision Making; Change Management

Note: the SIH-specified taxonomy contains 33 labels from the listed domains. We should encode them as prototype competency records and keep a source_reference field so future MoSPI/KCM alignment can be added transparently. [S1]

7. Competency scoring - exact approach

The model should never claim that an LLM can magically know an employee's real-world skill. PRAGYA estimates competency from multiple evidence sources.

Evidence	Prototype weight	Rationale
Diagnostic assessment	50%	Direct evidence of targeted knowledge
Recent assessment / quiz	20%	Recent evidence of understanding
Verified training evidence	15%	Shows participation but not mastery
Relevant experience	10%	Useful context but years alone do not prove mastery
Self-assessment	5%	Useful signal with lower reliability

Proposed formula: Score = 0.50 Diagnostic + 0.20 RecentAssessment + 0.15 Training + 0.10 Experience + 0.05 SelfAssessment. This is a prototype methodology created by the team, not an official MoSPI scoring policy.

Evidence drawer

Every visible competency score should be explainable through the evidence that contributed to it: diagnostic result, recent quiz, training evidence, relevant experience, and self-assessment. Store confidence separately from score.

8. Skill-gap engine

Basic gap: $\text{Gap}(c) = \max(\text{Target}(c) - \text{Current}(c), 0)$. Priority then incorporates role criticality, task relevance, organizational priority and recency so that the largest numeric gap is not automatically the most important one.

Skill	Current	Required	Gap	Example priority
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GIS	25	60	35	Critical
Python	40	70	30	High
AI/ML	20	40	20	High
SQL	75	60	0	No gap
Statistics	85	80	0	No gap

9. Recommendation engine

Use a hybrid recommender because a hackathon prototype will not have enough interaction data for reliable collaborative filtering. Rule-based scoring prioritizes the need; embeddings identify semantic relevance; metadata enforces fit.

Signal	Example	Role
Gap priority	GIS gap 35, critical	Decides urgency
Semantic similarity	Course text matches GIS/spatial analysis	Finds relevant content
Level fit	Beginner/intermediate course for current learner level	Avoids over/under-challenge
Coverage	Course covers 4 target competencies	Measures usefulness
Duration fit	2h course for available learning budget	Improves practicality
History	Already completed similar course	Avoids redundancy

Recommended prototype formula: $\text{CourseScore} = 35\% \text{ GapPriority} + 30\% \text{ SemanticSkillMatch} + 15\% \text{ LevelFit} + 10\% \text{ AssessmentCoverage} + 5\% \text{ DurationFit} + 5\% \text{ Novelty}$. Keep weights configurable.

10. iGOT and NSSTA/TPAC in PRAGYA

iGOT and NSSTA/TPAC should appear as distinct learning channels in the recommendation experience. The current CBC/Karmayogi ecosystem emphasizes role/competency alignment and mapping learning content to competencies. NSSTA currently publishes training offerings and TPAC-related documents. [S2] [S3] [S4] [S5]

Channel	PRAGYA treatment	Why distinct
iGOT	Digital course catalogue, course metadata, learning history and recommendation	Digital/self-paced ecosystem resource
NSSTA / TPAC	Structured training programme catalogue, eligibility, schedule and source	Institutional/training-programme intervention
PRAGYA virtual lab	Hands-on practice	Applies knowledge
PRAGYA assessment	Measurement intervention	Provides evidence for recalibration

Integration honesty

For the prototype, use a clearly labelled MOCK iGOT provider and a production-ready adapter interface. Do not claim a live government API unless authorized credentials and a working endpoint are actually available. A public KB-iGOT repository can be used as an implementation-pattern reference, not as a universal API contract. [S8]

11. RAG learning assistant

RAG means Retrieval-Augmented Generation. It lets the assistant find evidence in a learner's approved course material before generating an answer.

1. Upload PDF/PPT/video
2. Extract text or transcript
3. Chunk and attach page/slide metadata
4. Create embeddings
5. Store in Qdrant or pgvector
6. Retrieve top-k evidence for the question
7. Pass only relevant evidence + learner context to the LLM
8. Generate answer + source citation
9. Log source IDs for traceability

Example

Question: “What is stratified sampling?” PRAGYA retrieves the relevant pages from Survey Sampling.pdf, answers using that evidence, cites the page, and can connect the topic back to the employee's weak Sampling competency.

12. AI quiz and MCQ engine

The assessment engine is a pipeline rather than a single prompt.

Stage	Implementation
Content extraction	PyMuPDF / python-pptx / FFmpeg + faster-whisper
Chunking	Topic-aware text chunks with source metadata
Concept extraction	LLM/keyphrase extraction
Competency mapping	Match concepts to competency IDs
Question generation	LLM produces structured JSON
Distractors	LLM + rule checks for plausibility

Difficulty	Easy/Medium/Hard + optional Bloom level
Validation	Source support, ambiguity, single correct answer
Duplicate detection	Embedding similarity against existing questions
Traceability	Document, page/slide, chunk ID stored
Delivery	Adaptive quiz + instant result + explanation

13. Statistical virtual lab

Recommended first lab: Sampling Lab. It lets the employee choose population characteristics and sampling method, run a simulation, inspect estimates and sampling error, and answer a short reflection/assessment.

Lab step	Employee action	System output
Population	Select synthetic population	Dataset preview
Method	Choose random / stratified / cluster	Method explanation
Sample size	Set sample size	Constraint validation
Run	Execute sample	Estimate + variance
Inspect	View chart and benchmark	Sampling error/confidence interval
Reflect	Explain trade-off	Feedback
Assess	Answer 3-5 questions	Competency evidence

14. Employee dashboard - screen by screen

Screen	What it contains	What it helps the user do
Login	SSO-ready entry, secure login, language selection	Authenticates user and routes to learner view
Home	Competency summary, top gap, learning action, progress	Decides next action
Competency Passport	Domain tabs, radar, scores, confidence, evidence drawer	Understands capability and evidence
Skill Gaps	Current vs required bars, priorities, why gap matters	Chooses which gap to attack first
Learning Path	Timeline with courses, labs, assessments and explanations	Follows personalized sequence
Course Discovery	Search, filters, iGOT/TPAC tabs	Finds alternatives
Course Detail	Competencies, level, duration, prerequisites, source	Decides whether to start
AI Assistant	Contextual chat, citations, hints and practice prompts	Gets just-in-time help

Upload Material	PDF/PPT/video upload and processing status	Turns personal/course material into knowledge
Quiz Generator	Question count, difficulty, competency scope, source preview	Creates assessment
Quiz	MCQ interface, progress, adaptive difficulty indicator	Demonstrates learning
Results	Score, weak topics, explanations and source references	Understands errors
Skill Improvement	Before/after scores and evidence timeline	Sees competency change
Progress	Hours, completion, active learning path	Tracks learning activity
Notifications	Recommendation changes, due assessments	Keeps learner informed
History	Course/assessment timeline	Reviews prior learning

15. Administrator dashboard - screen by screen

Admin screen	Contents	Decision supported
Overview	Total employees, active learners, average readiness, open gaps, completion	Understand current workforce learning state
Competency distribution	Domain/skill distribution across workforce	Find broad capability strengths and weaknesses
Department heatmap	Departments x competencies	Target training where gaps cluster
Role analysis	Role requirements vs workforce capability	Check role readiness
Top gaps	Ranked organization-wide gaps	Prioritize programmes
Training effectiveness	Pre/post assessment trends and observed change	Evaluate interventions
Recommendation analytics	Frequently recommended content and catalogue gaps	Improve learning supply
Emerging skills	Trending competency gaps / demand signals	Prepare for changing needs
Predictive analytics	Prototype future skill demand projection	Support capacity planning
Employee drill-down	Authorized individual evidence view	Support targeted intervention
Integration health	Adapter status, freshness, errors	Monitor ecosystem connectivity

16. UI system and visual language

The UI should look like one product across every screen. The visual identity is deliberately dark, colourful and futuristic, but still professional enough for government technology.

Element	Design direction
Base	Midnight navy / indigo surfaces

Accent 1	Electric cyan / teal
Accent 2	Violet / magenta
Accent 3	Warm coral/orange for priority states
Cards	Glassmorphism, soft shadows, subtle glow
Charts	Neon gradients with restrained animation
Motion	Flowing competency lines, animated progress rings, soft node pulses
Typography	Clean sans-serif, high contrast
AI identity	Abstract glowing AI orb/waveform, no robot mascot
Employee illustration	One consistent recurring employee character style across learner screens
Accessibility	Do not rely on color alone; pair status colors with labels/icons

Consistency rule

Every generated screen, poster or prototype route should use the same PRAGYA wordmark, color system, card treatment, icon style and recurring employee visual. Do not mix light government-dashboard mockups with unrelated neon screens.

17. AI/ML stack and purpose

Component	Model/technique	Purpose	Why
Competency engine	Weighted evidence scoring	Estimate skill	Transparent and feasible
Embeddings	Multilingual E5/BGE family	Semantic matching	Good with small data
Vector DB	Qdrant / pgvector	RAG retrieval	Simple and scalable
LLM	Ollama + Qwen/Llama-family or approved API	Explanations, extraction, MCQs	Flexible local/API deployment
RAG	Retrieval + grounded generation	Course-specific assistant	Reduces unsupported answers
Recommendation	Rules + embeddings + metadata	Course ranking	Strong with limited interaction data
Adaptive assessment	Difficulty rules	Next-question selection	Simple and reliable
Forecasting	Regression/time-series trend model	Future skill demand	Explainable prototype
ASR	faster-whisper	Video transcription	Open-source practical choice

18. Data strategy

Type	Examples	Use
Official data	MoSPI/eSankhyiki datasets, OGD resources	Domain examples and labs
Official web resources	NSSTA offerings, TPAC documents, public iGOT course information	Course/programme metadata and grounding
Synthetic	Employees, roles, scores, learning histories	Workforce dashboard and demo without private HR data
Mock	iGOT API responses, enrolment/completion, SSO	Demonstrate integrations honestly

The current eSankhyiki portal and OGD catalogue provide public MoSPI statistical resources. Access to individual microdata can differ from access to aggregate/public datasets, so use public resources in the demo unless an official access path has been granted. [S6] [S7]

19. Database entities

Entity	Important fields
Users	id, username, password_hash, role, status, created_at
Employees	employee_id, user_id, department_id, role_id, education, experience, assignment
Departments	id, name, priority
JobRoles	id, name, description, career_level
Competencies	id, name, domain, description, source_reference
CompetencyRequirements	role_id, competency_id, required_level, criticality, priority
EmployeeCompetencies	employee_id, competency_id, score, confidence, updated_at
Courses	course_id, provider, title, description, duration, level, source_type
CourseCompetencies	course_id, competency_id, coverage_level
TrainingProgrammes	programme_id, provider, title, eligibility, schedule, source_reference
LearningHistory	employee_id, item_id, status, completed_at, score, hours
Assessments	id, title, type, competency_scope, material_id
Questions	id, assessment_id, text, options, answer, explanation, difficulty, competency_id, source_chunk_id
QuizAttempts	attempt_id, employee_id, assessment_id, score, started_at, completed_at
SkillScores	employee_id, competency_id, old_score, new_score, evidence_type, evidence_id, timestamp

Recommendations	employee_id, item_id, competency_id, source_type, score, reason, generated_at
UploadedMaterials	id, owner_id, filename, mime_type, size, status
Documents	id, material_id, page_slide, chunk_text, embedding_id, competency_ids
AIConversations	id, employee_id, query, response, source_chunk_ids, timestamp
Notifications	id, user_id, type, message, read_status
AuditLogs	id, user_id, action, resource, timestamp, metadata

20. Security and privacy

Control	MVP	Production direction
Authentication	JWT, hashed passwords	Authorized government SSO/OIDC/SAML
Authorization	Employee / Trainer / Admin roles	Fine-grained least privilege
File security	MIME/type/size checks, safe storage	Malware scanning, isolated processing, DLP
API security	Input validation, tokens	Gateway, rate limiting, mTLS where required
AI security	Prompt injection guardrails, source constraints	Policy engine, adversarial testing, model monitoring
Privacy	Synthetic demo data, minimization	Approved retention, access control, privacy governance
Audit	Login/upload/admin actions	Centralized immutable security logging
Encryption	HTTPS, protected secrets	Encryption at rest + managed keys

MeitY currently lists the Digital Personal Data Protection Rules, 2025, and CERT-In maintains Directions under Section 70B. Use these as production-readiness references, not as a claim that a student prototype is legally certified. [S9] [S10]

21. Integration architecture

Keep external ecosystems behind adapters so the core recommendation engine does not care whether data came from mock or production services.

Layer	Prototype	Future
Provider interface	search_courses(), get_course(), get_enrollment_status(), get_completion_status(), get_learning_history()	Same interface
iGOT adapter	Mock JSON API / seeded catalogue	Authorized production connector
NSSTA adapter	Curated programme dataset from public	Approved institutional data connector

	sources	
Identity	JWT login	Authorized government SSO
Data	Synthetic workforce	Approved enterprise HR/LMS integrations

22. MVP vs advanced scope

Priority	Features
MUST BUILD	Employee profile; competency framework; diagnostic assessment; evidence-backed score; skill-gap engine; hybrid recommendation; mock iGOT; NSSTA/TPAC catalogue; PDF RAG; AI assistant; MCQ generation; quiz; skill recalibration; learner dashboard; admin dashboard
SHOULD BUILD	Evidence drawer; recommendation explanation; adaptive difficulty; Sampling Lab; multilingual explanations; source-linked questions
IF TIME PERMITS	PPT ingestion; video transcription; future-skill forecast; competency graph; role readiness score
DO NOT BUILD NOW	Production iGOT credentials; full SSO deployment; enterprise HR integration; mobile app; blockchain; fine-tuned LLM; agent swarm; dozens of labs

23. Development order - start here

1. Create the core schema and seed 5 roles + 33 competencies.
2. Create one synthetic employee with known current and target scores.
3. Build the diagnostic assessment and competency scoring endpoint.
4. Build the skill-gap endpoint and prioritized gap UI.
5. Seed 20-30 mock iGOT courses and 10 NSSTA/TPAC programmes.
6. Build the hybrid recommendation service and “Why recommended?” explanation.
7. Connect the employee dashboard to the recommendation service.
8. Add PDF ingestion + RAG assistant.
9. Add MCQ generation + validation + quiz UI.
10. Add competency recalibration after quiz completion.
11. Add administrator analytics on the same data.
12. Then add virtual lab, multilingual, forecasting and extra polish.

First milestone

A user can log in, see their role, take a diagnostic, receive an evidence-backed competency profile, see a skill gap, receive iGOT/TPAC recommendations, ask a grounded question, take a generated quiz, see their

competency score update, and receive a changed recommendation.

24. Demo story

Time	Action	What evaluator should notice
0:00-1:00	Login + profile	Role-aware identity
1:00-2:00	Competency passport	Evidence-backed scoring
2:00-3:00	Gap analysis	Priority is role-aware
3:00-4:00	Learning path	iGOT + NSSTA/TPAC recommendations
4:00-5:00	AI assistant	RAG with source citation
5:00-6:00	Upload PDF -> MCQs	Source-grounded assessment
6:00-7:00	Quiz	Instant evaluation + weak topic
7:00-8:00	Score recalibration	Learning changes competency
8:00-9:00	Updated recommendations	Closed loop
9:00-10:00	Admin dashboard	Workforce intelligence

25. Judge questions - exact answers

Question	Answer
Why two dashboards?	Because SIH26101 explicitly asks for both learner and administrator dashboards. The learner needs a personal action view; the administrator needs an organization-wide workforce intelligence view. [S1]
Where is the AI?	In competency evidence processing, semantic matching, RAG, question generation, adaptive learning support and forecasting. AI is assigned to concrete problems rather than used as decoration.
How do you know an employee is skilled?	We do not claim certainty. We estimate competency from multiple evidence sources and keep confidence separate from score.
Is iGOT really connected?	The prototype uses an integration-ready adapter with mock data unless authorized production API access is available. We will not fabricate a live government API connection.
Why not collaborative filtering?	A hackathon prototype does not have enough user-item interaction data. A hybrid rules + embeddings approach is more reliable with small data.
Why RAG?	It grounds course-specific answers in approved learning materials and gives source citations.
Why is your solution unique?	The core differentiator is the closed loop: role requirements ->

	evidence-backed competency -> gap -> personalized learning -> assessment -> competency recalibration -> new recommendation.
What is synthetic?	Employee/workforce records and other demo-only data. Official statistical content is clearly separated from synthetic HR/workforce data.
Is the score an official government policy?	No. It is a prototype scoring method that can later be replaced or calibrated with an approved competency framework and assessment standard.
What would you build next after the hackathon?	Replace mock integrations with authorized services, strengthen competency calibration, integrate approved SSO and enterprise systems, harden security, expand labs and improve analytics.

26. Uniqueness comparison

Common baseline	PRAGYA
Login + dashboard	Role + competency context
PDF -> chatbot	RAG assistant tied to learner context and source citations
PDF -> 10 MCQs	Competency-mapped, difficulty-labeled, source-validated MCQs
Course list	Hybrid recommendation engine with explainability
Static skill score	Evidence-backed score + confidence + audit trail
Completed course = success	Assessment evidence changes competency
One dashboard	Separate learner and administrator views
Generic analytics	Workforce gaps, training effectiveness and future skill intelligence
Generic education app	Domain-specific statistical competencies and a Sampling Lab

27. What is official vs our enhancement?

Official SIH / ecosystem requirement	Our enhancement
Competency assessment	Evidence drawer + confidence model
Skill-gap analysis	Role-criticality + task-relevance prioritization
Personalized recommendations	Hybrid ranking + “Why recommended?”
iGOT integration	Adapter architecture + mock-provider demo
NSSTA TPAC recommendations	Distinct structured-programme track
AI MCQ generation	Source-traceability + quality gate
Learner dashboard	Animated competency visualization
Admin dashboard	Interactive heatmaps + drill-down

Adaptive learning	Closed-loop recalibration
Interactive / virtual learning	Sampling Lab
Predictive analytics	Explainable trend forecast

28. Final PRAGYA definition

Product statement

PRAGYA is an AI-powered competency-based learning and workforce intelligence platform for India's Official Statistical System that identifies role-specific competency gaps, recommends personalized training through iGOT and NSSTA/TPAC resources, supports source-grounded AI learning and automated assessment, and continuously recalibrates competency based on new evidence.

Primary user outcome: “I know what I need to improve and what I should do next.”

Administrator outcome: “I know where our workforce capability gaps are, which learning interventions appear effective, and what capabilities we may need next.”

Technical outcome: “A modular architecture separates competency intelligence, recommendation, RAG/LLM, assessment, analytics and external integrations so the hackathon MVP can evolve into a production-ready direction.”

29. Reference sources

These are the current public references checked for the project blueprint. Re-verify access conditions before final submission because government portals and integration credentials can change.

[S1] SIH26101 public problem statement / expected solution: <https://sih2026.vuce.in/ps/SIH26101>

[S2] Capacity Building Commission - Karmayogi Competency Model: <https://www.cbc.gov.in/karmayogi-competency-model-kcm>

[S3] iGOT Karmayogi - AI Driven Capacity Building Plan: <https://www.portal.igotkarmayogi.gov.in/training-pla-ai/>

[S4] NSSTA - Training Offerings: <https://nssta.gov.in/offerings>

[S5] NSSTA - Documents / TPAC: <https://nssta.gov.in/document>

[S6] MoSPI eSankhyiki: <https://esankhyiki.mospi.gov.in/>

[S7] Open Government Data - eSankhyiki catalogue: <https://www.data.gov.in/catalog/e-sankhyiki>

[S8] KB-iGOT public repository - integration contract reference: https://github.com/KB-iGOT/deterministic-chatbot/blob/main/docs/INTEGRATION_CONTRACT.md

[S9] MeitY - Digital Personal Data Protection Rules 2025: <https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa>

[S10] CERT-In - Directions 70B: <https://www.cert-in.org.in/Directions70B.jsp>